Collision lab

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Test momentum conservation with measured before-and-after velocities.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 4.*
The new force magnitude is 1/3 of the original because the same momentum change occurs over 3 times the time.

02. Worked example: explain the model
Small decrease may involve friction/external impulse or measurement error; no cause is established by the difference alone.

03. Read and interpret the model
Both +0.20 kg·m/s.

04. Apply the physics
Total 2 kg·m/s before and after; consistent with momentum conservation.

05. Explain your reasoning
Table must preserve raw data. Friction/track tilt; sensor noise or interval choice. Differences alone do not refute conservation.

06. Review: connect an earlier idea *Review: Unit 2, Day 11.*
4 s to top; 8 s total, assuming no drag and equal launch/landing heights.

07. Exit check
Opposite directions give opposite momentum contributions; unsigned addition misstates system momentum.

08. STRETCH: test the model
A interval -0.1 to 0.3 contains zero; B 0.6 to 1.0 does not.

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01. Warm-up: bring back an earlier idea *Review: Unit 3, Day 4.*
J=-12 N·s; first F≈-12 N; longer-time F≈-6 N. Both oppose the initial velocity.

02. Worked example: explain the model
Small decrease may involve friction/external impulse or measurement error; no cause is established by the difference alone.

03. Read and interpret the model
Both +0.20 kg·m/s.

04. Apply the physics
Initial total momentum = (0.50)(0.60) = +0.300 kg·m/s. Final total = (0.50 + 0.75)(0.23) = +0.2875 kg·m/s. Percent deviation = |0.2875 - 0.300|/0.300 × 100% ≈ 4.17%.

05. Explain your reasoning
Table must preserve raw data. Friction/track tilt; sensor noise or interval choice. Differences alone do not refute conservation.

06. Review: connect an earlier idea *Review: Unit 2, Day 11.*
Both ranges≈138.6 m. Times≈4 s and 6.928 s. The larger angle gives longer flight.

07. Exit check
5% deviation. Difference relative to initial is not an experimentally established uncertainty or probability.

08. STRETCH: test the model
Change from -0.4 to 0 kg·m/s; force -2 to 0 N. Bounds alone are not probabilities.